The Probability of a Lag Spike: Statistical Modeling of Network Latency

Moving Beyond "Average Ping" Using Right-Skewed Probability Distributions and Hypothesis Testing

Presenter

[Your Name / Group Members]

Course

Applied Probability & Statistical Infer

The Problem: Why "Average Latency" Lies to You

The mean can look excellent while the tail contains disruptive lag spikes.

WHAT PEOPLE ASSUME

Normal distribution

- Symmetric bell curve
- Tail risk can look too small

REALITY

Right-skewed latency

- Hard lower bound around 5–10 ms
- Long right tail; Log-Normal / Gamma fit be

TAKEAWAY **Average latency is only one part of network quality; tail probability measures disruption risk.**

Probability of a Lag Spike

Why Does This Matter? The Impact of Tail Latency

Real-time applications are sensitive to rare but disruptive spikes.

1 Video Conferencing

Zoom / Teams

- Jitter and lag spikes
- Frozen frames, dropped audio

2 Esports & Cloud Gaming

Consistency matters

- 100 ms+ spikes can decide outcomes
- Low averages do not guarantee stability

3 Financial Operations

Time-sensitive processing

- Real-time transactions
- Deterministic latency bounds matter

4 Consumer Advocacy

ISP SLA auditing

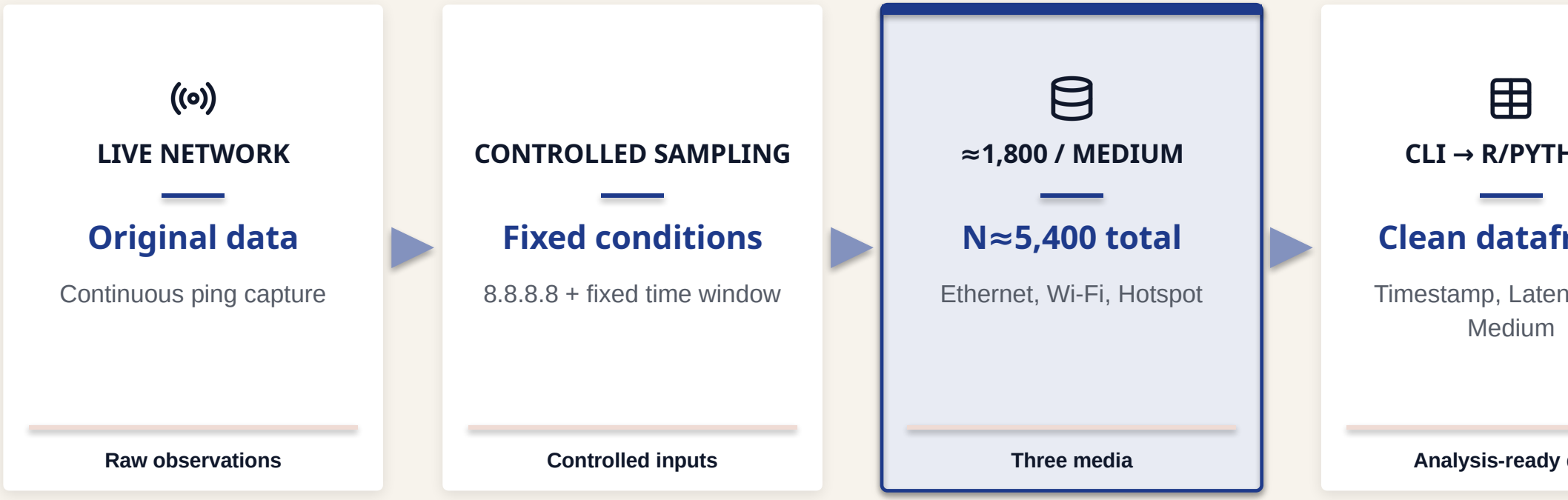
- Measure reliability statistically
- Tail metrics complement averages

TAKEAWAY **Tail latency turns network quality into a measurable probability-of-disruption question.**

Probability of a Lag Spike

Primary Data Collection: Empirical Network Sampling

Original observations are collected from live networks through a reproducible pipeline.



TAKEAWAY **Controlled sampling keeps the measurement process reproducible while comparing network media.**

Statistical Methods: From Data to Inference

The workflow combines exploration, distribution fitting, hypothesis tests, and uncertainty estimation.

1 · Explore

EDA (Exploratory Data Analysis)

- Boxplots, violin plots
- Density, skewness, kurtosis

2 · Fit

Distribution modeling

- Log-Normal, Gamma, Weibull
- MLE via fitdistrplus

3 · Test

Goodness-of-fit & inference

- K-S, Chi-Square, AIC/BIC
- ANOVA or Kruskal-Wallis + Tukey HSD

4 · Estimate

Uncertainty

- 95% Bootstrap CI: median
- 95% Bootstrap CI: 99th percentile

TAKEAWAY **The analysis progresses from describing latency to modeling its distribution and quantifying uncertainty**

Beyond the Mean: Calculating the Probability of Disruption

Define a lag spike at T = 100 ms and evaluate P(X > 100) = 1 − F_X(100).



Project-stated illustrative estimates; final values should be replaced with fitted empirical results.

TAKEAWAY **At T = 100 ms, estimated disruption risk rises from ≈0.1% on Ethernet to ≈12.0% on Hotspot.**